

SHAKYA MARASINGHA

Materials Science & Engineering Undergraduate

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[shakya-22 \(Shakya Marasingha\)](#)

PROFESSIONAL SUMMARY

Highly motivated Materials Science & Engineering undergraduate at the University of Moratuwa with strong interest in advanced materials, nanotechnology, electrochemical systems, energy storage, and sustainable engineering. Experienced in interdisciplinary research involving mathematical modelling, computational analysis, reaction kinetics, electrochemistry, and quantum-inspired materials science. Currently building skills in machine learning and MATLAB to complement engineering and research work. A proactive team player with leadership experience in IEEE and university organizations, seeking internship and research opportunities to further develop technical and professional expertise.

EDUCATION

Bachelor of the Science of Engineering (B.Sc. Eng.) — Materials Science & Engineering | University of Moratuwa

March 2024 – Present

- Relevant coursework: Materials Characterization, Nanomaterials, Electro-materials, Electrochemical Systems, Materials Thermodynamics, Materials Kinetics, Functional Materials, Sustainable Engineering

Diploma in Information Technology | ESOF Metro Campus

April 2023 – May 2024

Diploma in English Language and Literature | BCI Campus

April 2023 – December 2023

WORK EXPERIENCE

General / Operations Intern | Bank of Ceylon (BOC)

2023 – 2024

- Provided customer service support, assisting clients with routine banking queries and transactions
- Supported marketing initiatives to promote bank products and services
- Assisted with account handling and account-related administrative tasks
- Supported front-office and back-office staff with financial transactions and record-keeping.
- Developed practical exposure to the banking and financial services sector, strengthening attention to detail and professionalism.

ENGINEERING PROJECTS

Artificial Photosynthesis: From Biological Blueprint to Quantum Engineering | University of Moratuwa — MT2181,

Solid State Materials

- Investigated quantum mechanical principles governing artificial photosynthesis
- Studied semiconductor photocatalysts and electronic band structures
- Analysed charge separation, electron transport, and photocatalytic reactions
- Explored Z-scheme mechanisms, water oxidation, hydrogen evolution, and quantum efficiency
- Integrated concepts from quantum mechanics, materials science, electrochemistry, and renewable energy

Kinetic Modelling of Dopamine Transport in the Synaptic Cleft | University of Moratuwa — MT2230, Kinetics of

Materials

- Developed a mathematical model describing dopamine transport using Poisson–Nernst–Planck equations
- Incorporated diffusion, electro-diffusion, Michaelis–Menten kinetics, and receptor binding models
- Investigated physiological and pathological neurotransmitter transport
- Applied computational and analytical methods to analyse dopamine concentration profiles

LEAP Laptop Stand | Engineering Design & Workshop Project

- Designed and developed an ergonomic laptop stand prototype with adjustable design, cooling-perforated wooden platform, natural tea-waste thermal insulation, and memory-foam cushioning
- Applied sustainable and user-centred design principles throughout

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Team Lead, Publicity Committee (Octwave 3.0) | IEEE Student Branch, University of Moratuwa

April 2026 – Present

- Lead publicity activities and coordinate promotional campaigns
- Manage team collaboration and event communications

Team Member, Publicity Committee (Moraforesight 4.0) | IEEE Student Branch, University of Moratuwa

April 2026 – Present

Team Member, Publicity Committee | IEEE NPSS Student Branch Chapter

April 2026 – Present

Co-Chair, Energy Market Webinar Session | IEEE PES Student Branch Chapter

March 2026 – April 2026

- Coordinated webinar planning and execution
- Assisted with speaker communication and event management

Team Member, Organizing Committee | IEEE PES Student Branch Chapter

February 2026 – Present

Marketing Team Member & Finance Panel Member | Mora Lenz

February 2026 – Present

- Hosted the team for year-long sponsorship programme

Team Member, Editorial Committee | SOMES

July 2026 – Present

Student Member, Institution of Engineers – Sri Lanka

July 2024 – Present

- Currently holding the membership in IESL – membership no. – S33302

Committee Member | Classical Music Society, University of Moratuwa

October 2025 – Present

VOLUNTEERING

Team Member, Arts & Culture | Rotaract Club of University of Moratuwa

April 2025 – May 2025

- Participated in community engagement activities while developing teamwork and leadership skills

TECHNICAL SKILLS

Engineering: Materials Science, Materials Characterization, Nanomaterials, Electrochemical Systems, Functional Materials, Sustainable Engineering, Materials Kinetics, Solid-State Physics

Research: Scientific Literature Review, Mathematical Modelling, Computational Analysis, Reaction Kinetics, Data Interpretation, Technical Report Writing

Software & Programming: AutoCAD, Solid-Edge, ImageJ, Python, C#, MATLAB (Academic, in progress), Machine Learning (in progress), Microsoft Excel, Microsoft Word, Microsoft PowerPoint

Professional Skills: Analytical Thinking, Research Skills, Leadership, Teamwork, Problem Solving, Project Coordination, Communication, Time Management

PROFESSIONAL AFFILIATIONS

- IEEE Student Branch – University of Moratuwa
- IEEE Nuclear & Plasma Sciences Society (NPSS)
- IEEE Power & Energy Society (PES)
- Rotaract Club of University of Moratuwa
- SOMES
- Classical Music Society – University of Moratuwa
- AIESEC CS

LANGUAGES

English — Professional Working Proficiency | Sinhala — Native Proficiency

